

Project Documentation

Customer Feedback Intelligence & Recommendation System

(End-to-End Data Science + NLP + BI Project)

1. Project Overview

This project focuses on analyzing customer feedback using Natural Language Processing (NLP) and Machine Learning techniques to extract meaningful insights and provide actionable business recommendations.

Customer reviews contain valuable information about product quality, delivery experience, and overall satisfaction. However, manually analyzing large volumes of feedback is inefficient.

This system automates the process by:

- Classifying sentiment (Positive / Negative)
- Detecting key issues
- Recommending business actions
- Visualizing insights through dashboards
- Providing real-time predictions via API

2 Problem Statement

Organizations receive thousands of customer reviews daily. The key challenges include:

- Identifying customer satisfaction levels
- Detecting common issues
- Understanding product performance
- Taking data-driven decisions

Manual analysis is time-consuming and error-prone.

This project aims to solve this problem using an automated system.

◆ 3. Objective

- Analyze customer feedback using NLP
 - Classify sentiment (Positive / Negative / Neutral)
 - Detect common issues (delivery, quality, payment, etc.)
 - Build a machine learning model for sentiment prediction
 - Generate actionable business recommendations
 - Store and manage data using MySQL
 - Create an interactive Power BI dashboard
 - Develop a Flask API for real-time predictions
-

◆ 4. Tech Stack

Category	Tools Used
Programming	Python
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
NLP	TF-IDF, Text Preprocessing
Database	MySQL
Visualization	Power BI
Backend	Flask API
Tools	Jupyter Notebook, PyCharm, Postman, GitHub

5. Project Architecture

Customer Reviews Dataset



MySQL Database



Python (EDA + NLP + ML)



Sentiment + Issue + Recommendation



Power BI Dashboard



Flask API (Real-time Prediction)

◆ 6. Project Phases

◆ Phase 1: Problem Understanding

Understanding business needs:

- What do customers feel?
 - What are the major complaints?
 - What improvements are required?
-

Phase 2: Data Collection

Dataset includes:

- Review Text
 - Rating
 - Product Name
 - Category
-

Phase 3: MySQL Database Design

Tables:

1. Reviews Table

- review_id
- product_name
- review_text
- rating

2. Results Table

- review_id
- sentiment
- issue

3. Action Table

- issue
 - recommendation
-

Phase 4: Data Cleaning

Performed using Python:

- Removed duplicates
 - Handled missing values
 - Standardized text
 - Converted to lowercase
-

Phase 5: Exploratory Data Analysis (EDA)

Key insights:

- Rating distribution
- Positive vs Negative reviews
- Product-wise feedback
- Common complaint patterns

Visualizations:

- Bar charts
 - Pie charts
 - Count plots
-

Phase 6: NLP Preprocessing

Steps:

- Text cleaning
 - Tokenization
 - Stopword removal
 - TF-IDF vectorization
-

Phase 7: Machine Learning Model

Model used:

- Logistic Regression

Target:

- Sentiment classification

Evaluation:

- Accuracy
 - Confusion Matrix
 - Classification Report
-

Phase 8: Issue Detection & Recommendation

Rule-based logic used:

Issue Detection:

- “late” → delivery issue
- “bad” → quality issue
- “refund” → payment issue

Recommendation Engine:

- Delivery issue → Improve logistics
- Quality issue → Improve product quality
- Payment issue → Improve refund process

Phase 9: Model Saving

Saved using pickle:

- model.pkl
- vectorizer.pkl

Phase 10: Power BI Dashboard

KPIs:

- Total Reviews
- Positive Reviews %
- Negative Reviews %
- Average Rating

Visuals:

- Sentiment distribution
 - Top issues
 - Product performance
 - Monthly trends
-

Phase 11: Flask API

Input:

```
{  
  "review": "Delivery is late and product is bad"  
}
```

Output:

```
{  
  "sentiment": "Negative",  
  "issue": ["delivery", "quality"],  
  "recommendation": "Improve delivery and product quality"  
}
```

Phase 12: Deployment & Testing

- API tested using Postman
- Dashboard created in Power BI
- Code structured and uploaded to GitHub

7. Key Results & Insights

- Identified major issues affecting customer satisfaction
 - Automated sentiment classification
 - Generated actionable recommendations
 - Reduced manual effort in feedback analysis
 - Improved decision-making process
-

8. Learning Outcomes

- Practical experience with NLP and ML
 - Understanding of real-world data pipelines
 - Hands-on experience with MySQL integration
 - Dashboard creation using Power BI
 - API development using Flask
-

9. Solutions Implemented

- Applied TF-IDF for better feature extraction
 - Used structured pipeline for data flow
 - Implemented rule-based logic for issue detection
 - Ensured proper data preprocessing
-

10. Folder Structure

Customer-Feedback-Intelligence-System/

```
├── data/  
├── notebooks/  
├── sql/  
├── models/  
└── app/
```

- └─ dashboard/
- └─ reports/
- └─ README.md
- └─ requirements.txt

5. Conclusion

This project successfully demonstrates how customer feedback can be transformed into meaningful insights using data science techniques.

The system not only analyzes sentiment but also identifies key issues and provides business recommendations, making it highly valuable for real-world applications.
